Syllable token frequencies reveal a high cross-linguistic prevalence of the Sonority Sequencing Principle

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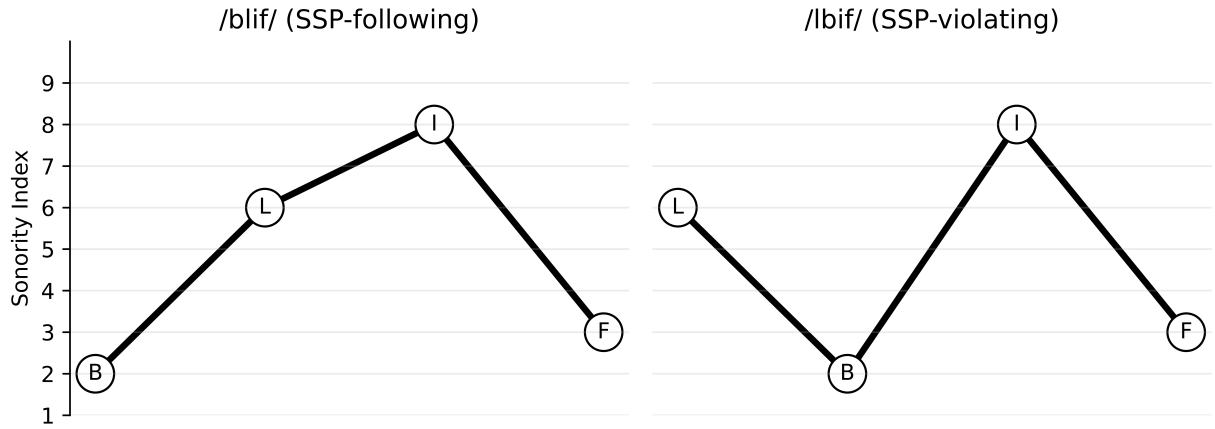
Syllable token frequencies reveal a high cross-linguistic prevalence of the Sonority Sequencing Principle

Sonority is a phonological feature that describes the perceptual salience of a given phoneme relative to neighbouring ones (Blevins, 1995; S. Parker, 2008; S. G. Parker, 2002; Selkirk, 1984). While the acoustic correlates of sonority are debated, some physical features such as intensity have been shown to correlate strongly with sonority indices in spoken languages (Fant et al., 2000; Jany et al., 2007; S. Parker, 2017; S. G. Parker, 2002; but see Albert & Nicenboim, 2022 for a different perspective). Phonemes can be hierarchically organised according to their sonority, with those at the lower end having lower sonority and those at the higher end having higher sonority (e.g., Obstruents < Nasals < Liquids < Glides < Vowels, Clements, 1990). This hierarchy is often represented as a numerical scale (e.g., 1 for lowest sonority, 9 for highest sonority, S. Parker, 2008). Based on this sonority hierarchy, some syllables are considered well-formed across languages while others are marked as ill-formed.

The Sonority Sequencing Principle (SSP) is a phonotactic rule that captures how sonority constrains the ordering of phonemes within a syllable. The SSP states that each syllable should display a unique peak of sonority at the nucleus, and that sonority should rise toward the nucleus and fall away from it (S. G. Parker, 2002; Selkirk, 1984). Syllables are classified as following or violating the SSP by examining the sequence of sonority values assigned to each of its phonemes Figure 1.

Figure 1

Schematic representation of the sonority hierarchy. The syllable /blif/ (left panel) has a sonority contour of 2-6-8-3 (8 being the vowel /i/, nucleus of the syllable) and is therefore considered well-formed, whereas /lbif/ (6-2-8-3, right panel) is considered ill-formed because the onset displays a sonority dip from the first to the second phoneme, violating the progressive increase in sonority that characterizes well-formed syllables.



Experimental evidence suggests that the SSP plays a pivotal role in processing spoken (Berent et al., 2007, 2011; Berent & Lennertz, 2010; Miozzo & Buchwald, 2013; S. Parker, 2017) and sign languages (Perlmutter, 1993; Sandler, 1993; Stone et al., 2018). Studies on newborns, infants, and adults show a consistent preference toward well-formed syllables (SSP-following) over ill-formed ones (SSP-violating), even when well-formed syllables are unattested in the participants' native language. For example, English-native adults prefer /bn/ (SSP-following) over /lf/ (SSP-violating) at syllable onset, although neither consonant cluster occurs in English (Berent et al., 2011; Berent & Lennertz, 2010). Few-day-old newborns show different brain haemodynamic responses when listening to SSP-following or violating syllables (e.g., *bran* vs. *lbif*) suggesting that precursors of the SSP are already present at birth (García-Castro et al.,

2026; Gomez et al., 2014). Such preferences for sonority-compliant syllables, even with unattested materials or very limited language experience (as in newborn infants) have been considered supporting evidence for the status of the SSP as a linguistic universal with which humans might be equipped from birth.

The universal status of the SSP is, however, still a matter of ongoing debate. Violations of the SSP are not uncommon across languages, with examples including fricative-stop consonant clusters at syllable onset (e.g., /ʃt/, as in the German word *Stein*) and their reverse counterpart at coda, stop-fricative clusters (e.g., /ks/, as in the English word *clicks*) (Kawasaki-Fukumori et al., 1997; Morelli et al., 2003). Languages that generally allow considerably complex consonant clusters at syllable onset tend to show SSP violations more frequently. This for instance the case of Polish, which allows for clusters of up to three consonants at syllable onset (e.g., *chrząszcz* /ˈxʂɔ̃wʂt͡ʂ/ [beetle]) (Gussmann, 2007; Jarosz, 2017), or Georgian (e.g., მწვრთნელი /mt͡sʼvrtʰneli/ [trainer]) (Crouch et al., 2023).

A handful of studies have provided a quantitative approach toward the cross-linguistic prevalence of the SSP. Evidence of a preference for SSP-following consonant clusters across 43 languages was reported, with type frequencies of 97 distinct consonant clusters reflecting an overall 72% compliance with the SSP (Frisch, 2015). Type frequencies capture the number of distinct items (*types*, such as consonant clusters or syllables) in which a linguistic pattern (e.g., SSP-following structures) is present. When type frequencies were analyzed across 496 languages from 58 language families, it was found that 32% of the consonant clusters at word onsets violated the SSP, compared with 30% at word-final position (Yin et al., 2023). Both studies relied on type frequencies of attested consonant clusters in a sample of languages, counting the number of consonant clusters that followed or violated the SSP in each of those

languages, or simply establishing the presence or absence of SSP violations in them. While this approach provides useful information about languages' tolerance of SSP violations (or lack thereof), it does not provide a direct estimate of how often speakers of such languages encounter SSP violations in natural speech. This is because type frequency does not necessarily correspond on a one-to-one basis to its rate of occurrence in natural speech. Consider an imaginary language that allows for 20 consonant clusters in total, one of which is /st/. While /st/ represents a sizable portion of the language's phonotactical repertoire (5%), how frequently speakers encounter /st/ depends on at least two factors: the number of words containing /st/, and the frequency of each of those words. If /st/ occurs in only a few words or only in low-frequency words, its share in naturalistic speech would be lower than suggested by the language's phonological repertoire.

Another pitfall is that the phonological transcriptions used in most studies are drawn from lexical inventories and often lack syllable boundaries (Yin et al., 2023), making it difficult to draw reliable conclusions about the prevalence of the SSP in natural speech. The lack of syllabic boundaries results in an overepresentation of word-initial and word-final syllables, while word-medial consonant clusters remain unobserved, leaving a substantial portion of the syllabic structures uttered in natural speech unaccounted for.

In the present study, we aim to overcome these two pitfalls by providing a quantitative estimate of the cross-linguistic prevalence of the Sonority Sequencing Principle (SSP) that takes into account the frequency with which speakers encounter SSP-following and SSP-violating structures in natural language. To this end, we considered syllabic token frequency, which is a measure of how often a particular syllable is uttered per million words in natural speech in a given language, and has been

shown to be relevant in language processing and production tasks (Cholin & Levelt, 2009; Duchon et al., 2013; Perea & Carreiras, 1998). Additionally, to extract the syllabic repertoire of each language in the dataset, we used a large sample of syllabified phonological transcriptions that include word-medial syllables, which were not always represented in previous work. We then used a Bayesian modelling approach to estimate the prevalence of the SSP in a large dataset comprising 43,229 syllable types across 67 languages, yielding estimates of the proportion of SSP-following syllables in each language adjusted for the relative frequency of each syllable in natural speech.

Methods

Raw data, reproducible code, and project documentation are available in the Open Science Framework (https://osf.io/yeqms) and GitHub (https://github.com/NeuroDevCo/ssp-corpus).

Data collection and preprocessing

Phonological transcriptions were scraped from word entries in the Wiktionary database. Wiktionary is an open-source, free database containing 10,050,751 lexical entries from 8,244 languages contributed by 4,463,976 users (as of June 10th, 2026), many of which include at least one phonological transcriptions for a given word. The wikipron Python package (version 2.2.0, Lee et al., 2020) was used to scrape all available phonological transcriptions of words for all 7,859 languages included in the Glottolog database (Nordhoff & Hammarström, 2011). This resulted in approximately 1.7 million phonological transcriptions across 900 languages. Languages with less than 100 words were excluded, resulting in a final dataset of 171,057 words from 77 languages.

Available lexical frequencies for each word in each language were extracted using the *wordfreq* Python package (Speer, 2022), which provides an interface for retrieving lexical frequency data for words in over 40 languages. When lexical

frequencies were not available for a particular language (e.g., the ISO 639-3 code did not match any language available in *wordfreq*), the closest equivalent language was used, if available. Equivalent languages were identified based on their distance from the original language according to the CLDR v38.11 standard[1]. The rationale for using equivalent languages to retrieve missing lexical frequencies was that users of close languages are likely to use similar words to refer to the same objects, and lexical frequencies were therefore expected to be reasonably comparable across equivalent languages. Words with missing lexical frequency scores were excluded from the dataset.

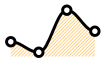
For the phonological transcription of each word, annotations about tonal features were stripped, then each phonological transcription was segmented into syllable candidates. When possible, syllable boundary markers were used, either explicit (".") or implicit ("ˈ" for primary stress, "ˌ" for secondary stress). In the absence of syllable markers, the word was assumed to be monosyllabic. Segments of each syllable candidate were then classified as consonants (C), vowels (V), or semi-vowels (v). Individual vowels, consecutive strings of vowels and semi-vowels (e.g., V, VV, Vv, VVv) were classified as vocalic clusters. Syllable candidates with more than one vocalic cluster (e.g., CVvCV, VCV) were flagged as multisyllabic and excluded from the dataset. This methods allowed to extract word-medial syllables when possible. Duplicated syllables within each language were removed (e.g., if the syllable /ka/ appeared in the Spanish words /ˈka.sa/ and /kaˈpo/, it was counted only once in the Spanish dataset). This procedure yielded one syllabic repertoire for each language.

[1]The CLDR system assigns a numeric value to each pair of language codes based on whether both codes are assigned to identical languages with different names (e.g., Norwegian Bokmål → Norwegian, score 0), minor regional differences (score 1-3, e.g., Norwegian Bokmål → Norwegian), and languages with significant but unproblematic regional differences (score 10-24, e.g., Afrikaans → Dutch, Wu Chinese → Mandarin Chinese), etc.

To evaluate whether each extracted syllable followed or violated the SSP, a sonority score was assigned to each phoneme on a scale from 1 to 9, using the *panphon* Python package (version 0.22.2, Mortensen et al., 2016). Syllables with more than one peak (i.e., exhibiting a local minimum in the sonority contour) were classified as SSP-violating, and as SSP-following otherwise. Following this rule, syllables with a flat sonority contour (e.g., English word sphinxs /ˈsfɪŋks/ in which the first two phonemes bear identical sonority index value, 3-3-8-5-1-3) were classified as SSP-following. Whether consecutive segments with same sonority index comply or not with the SSP is a matter of debate (e.g., Tropf, 1987; Berent et al., 2007; Eckman, 1987; Santolin et al., 2023), and this reason we report complementary analyses in which flat sonority contours were classified as SSP violating, showing equivalent results (see Appendix C).

Table 1

Examples of SSP-following and SSP-violating syllables in the dataset. Sonority index contours are shown for each syllable numerically and graphically. For example, the syllable /ˈskul/ in Assamese shows a violation at onset, as sonority drops from /s/ (sonority index 3) to /k/ (sonority index 1) before rising again at nucleus /u/ (sonority index 8).

Language	Syllable	Original word	Sonority index	
SSP-following				
Egyptian Arabic	/dˤaːd/	ضـ| /dˤaːd/	2-9-2	
Modern Greek	/bros/	γαμπρόσ| /ɣamˈbros/	2-6-9-3	
Finnish	/dɑ̪u̯n/	touchdown| /ˈt̪ɑ̪tsˌdɑ̪u̯n/	2-9-7-5	
Hindi	/pʰɛ/	फ़हम| /pʰɛ.ɦɛ̃m/	1-9	
Russian	/font/	фонт| /font/	3-9-5-1	
SSP-violating				
Egyptian Arabic	/nisr/	نسر| /nisr/	5-8-3-6	
Modern Greek	/ftis/	συγγραφτείσ| /si.ɣraˈftis/	3-1-8-3	
Finnish	/s̪trø̪m/	ångström| /ˈo̪ŋ.s̪trø̪m/	3-1-6-9-5	
Hindi	/sʈɛ̃n/	स्तन| /sʈɛ̃n/	3-1-9-5	
Russian	/fpɐ́ɬ/	вполглаза| /fpɐ́ɬˈgɬazə/	3-1-9-6	

To calculate the frequency of each syllable, we summed the lexical frequency (counts per million words) of all words in the dataset which contained that syllable (see Duchon et al., 2013 for a similar approach). The rationale behind this approach is that,

whenever a word is uttered, all of its syllables are necessarily uttered as well. Therefore, summing the lexical frequencies of all words containing a given syllable yields an estimate of that syllable's frequency in natural speech. The final dataset comprised 43,229 syllables from 67 languages from 43,229 distinct language macro-families.

Data analysis

We used a multilevel Bayesian regression model to estimate the probability of each syllable in the dataset following (or violating) the SSP, considering the language it belonged to and its syllabic frequency. Bayesian models estimate a probability distribution for each parameter in the model, known as the *posterior* distribution. This is obtained by combining prior knowledge about the parameter (specified as a *prior* distribution based on domain knowledge) with the observed data via the Bayes rule. The posterior distribution provides inferential information about how likely each parameter is to be equal or close to zero. To estimate the posterior distribution of the model we used Hamiltonian Monte Carlo (HMC), an algorithm that explores the posterior distribution iteratively, as implemented in *PyMC3* (version 6.2.0, Patil et al., 2010) with four chains of 2,000 iterations, 1,000 of which were warm-up iterations.

We modelled the response variable y_{ij} (1 if the syllable was SSP-following, 0 otherwise) using a Bernoulli distribution with probability parameter θ_{ij}. We specified a linear regression model for θ_{ij} which included fixed regression coefficients for the intercept (α) and the slope of syllabic frequency (β), and random effects u_j and w_j for each fixed coefficient, respectively. We specified weakly informative priors for all parameters in the model (see Equation 1 in Appendix A).

We report the median and 95% credible interval (CrI) of the posterior distribution for the parameters of interest. The 95% CrI can be interpreted the interval that has 95% probability of containing the true value of the parameter (Kruschke & Liddell, 2018).

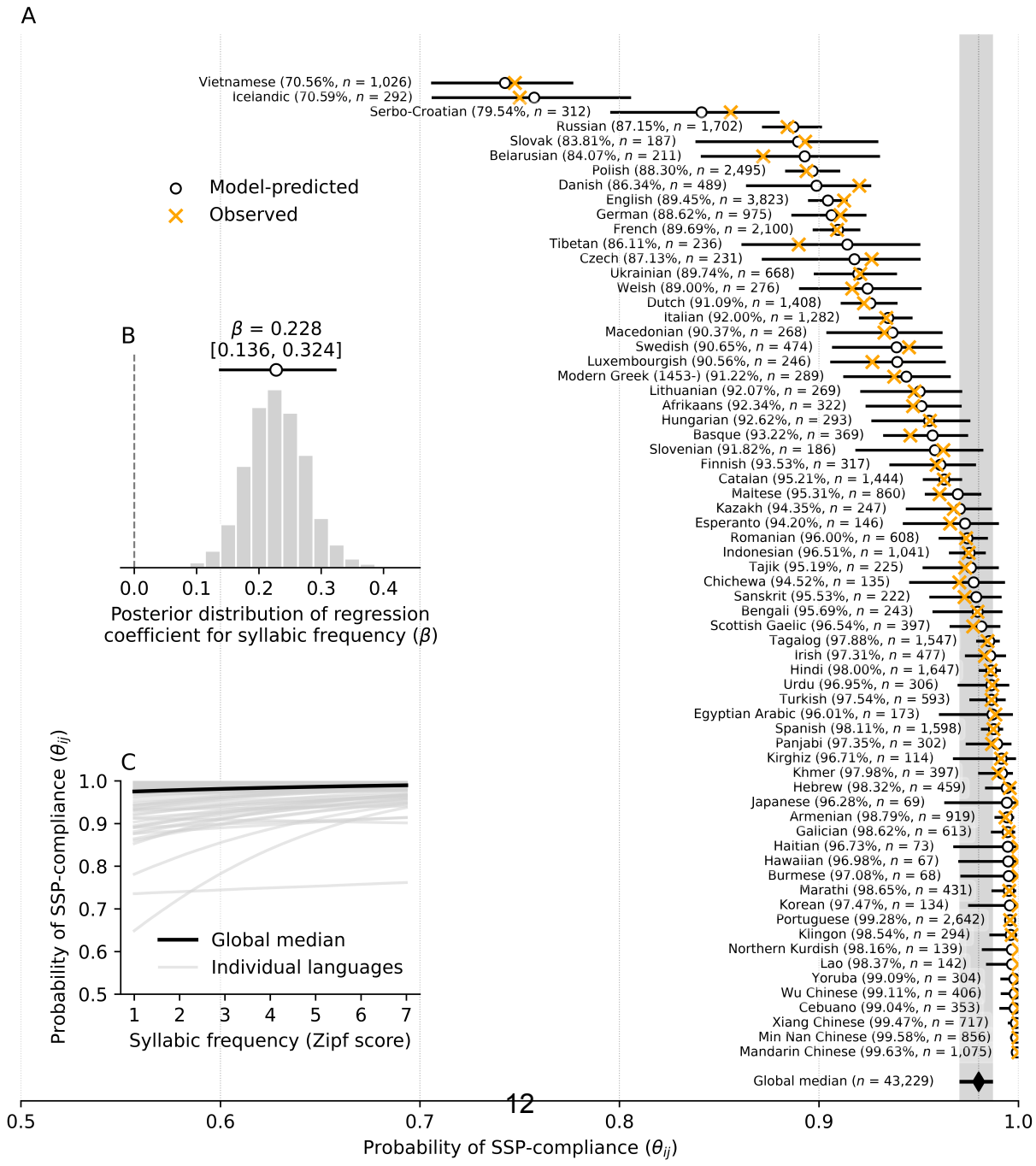
Parameters whose 95% CrI excluded zero were concluded to be reliably different from zero.

Results

The global proportion of SSP-following syllables across languages was θ = 0.980 (95% CrI = [0.970, 0.987]), and the prevalence of the SSP for each language ranged from 74.259% (Vietnamese, 95% CrI = [70.559%, 77.686%]) to 99.915% (Mandarin Chinese, 95% CrI = [99.635%, 99.990%]) (see Figure 2). Only 2.99% of the languages in the sample showed less than 80% of SSP prevalence, 11.94% languages showed less than 90% of SSP prevalence, and 31.34% languages showed less than 95% of SSP prevalence. Syllabic frequency showed a positive association with the probability of SSP compliance (β = 0.228, 95% CrI = [0.136, 0.324]), indicating that SSP-following syllables were, on average, more frequent than SSP-violating syllables.

Figure 2

A) Observed and model-predicted SSP prevalence (median and 95% CrI). B) Posterior distribution of the regression coefficient for syllabic frequency (β). More positive or negative values show a more positive or negative association between this predictor and SSP compliance (logit scale). C) Model-predicted of SSP compliance as a function of syllabic frequency. Growing lines indicate a positive association between both variables.



Discussion

Previous experimental studies had pointed to the Sonority Sequencing Principle (SSP) as a universal constraint defining the structure of syllables, key units in speech processing (e.g., Berent & Lennertz, 2010; Berent et al., 2011; García-Castro et al., 2026; Gomez et al., 2014). However, it remained unclear to what extent well-formed syllables according to the SSP are actually more common than ill-formed ones across languages. The present study addressed this issue by quantifying the prevalence of the SSP across a diverse set of languages, using syllabified phonological transcriptions that include all syllables composing words (not just word-initial or final ones) while also accounting for how often syllables are uttered in natural speech. Results revealed that SSP-following syllables are, on average, significantly more frequent than SSP-violating ones across the languages in our dataset, with an overall cross-linguistic conformity of 98.01%. In addition, the majority of languages in our sample followed the SSP more than 90% of the time.

A key novel aspect of the current study compared to prior work is the use of syllabic token frequencies (instead of type frequencies) which more closely reflect how often speakers actually encounter SSP-following or SSP-violating syllables in everyday speech. If SSP-following syllables occur more often than SSP-violating ones, estimates of SSP -prevalence likely underestimate their true value when syllabic token frequencies are not incorporated into the analysis. This may partially explain the higher prevalence of SSP compliance found here compared to other studies [e.g., 72% of SSP-following consonant clusters in 47 languages, Frisch (2015), see also Watson (2002); Gishri (2009); Al Tamimi and Al Shboul (2013); Yin et al. (2023); vs. 98.01% in our study]. Consistent with this possibility, we found a positive, non-zero association between syllabic token frequency and the probability of SSP compliance, suggesting

that SSP-following syllables tend to be uttered more frequently in natural speech. Taken together, our results suggest that most languages in our dataset comply with the SSP, albeit to varying degrees, and that violations of the SSP are exceptional for the majority of languages.

Complementarily, we also found that SSP violations, while uncommon, are attested in most languages in our dataset (see Yin et al., 2023 for a similar pattern of results). There are a few methodological considerations to keep in mind when interpreting these findings. First, syllables included in our dataset were not randomly sampled. Instead, each syllable was included based on the availability of at least one word containing that syllable in the corresponding language in Wiktionary, which may have introduced some degree of selection bias into our dataset (see Yin et al., 2023 for a similar patter of results). Second, while the present study includes multiple languages from a range of macro-families, the number of observations available for some languages is reduced, especially when comparing Indo-European languages with those from other families. This unequal data availability across languages is, however, partially accounted for by the partial pooling of random effects in multilevel models (see Gelman et al., 2007 for a detailed explanation).

Our findings can be interpreted within the Optimality Theory (OT) framework (Frisch, 2015; Prince & Smolensky, 2004), according to which linguistic universals are not hard, binary rules but rather general biases towards certain type of structures that most languages favour. Within this framework, exceptions to the SSP can be understood as the result of the SSP interacting with other phonotactic rules that speakers of certain languages weigh more strongly. Under OT, sonority constraints on syllabic structure remain useful for generating predictions about the well-formedness of

syllables, despite the cross-linguistic variability in the degree of compliance with this principle (Dell & Elmedlaoui, 1985).

Previous experimental work had capitalized on cross-linguistic differences in SSP compliance to investigate the role of sonority constraints on speech perception. For instance, Russian native adults correctly identified SSP-violating syllables (e.g., *lbif*) as monosyllabic, whereas English native adults misperceived them as a disyllabic (e.g., *lebif*) splitting the ill-formed consonant cluster /lb/ in two syllables (/le/ and /bif/, Berent et al., 2007). These results were attributed to Russian speakers' more frequent exposure to SSP-violating syllables compared to English speakers. However, contrary to one might expect, Russian still shows an overall higher prevalence of SSP-following structures despite its lower compliance with the SSP compared to other languages (87.45% vs. 90.45% in English, for example). Similar results were found in other Slavic languages such as Polish (88.72%), Ukrainian (91.97%), and Czech (91.78%), in which SSP violations are traditionally known to be more frequent than in other language families. A possible explanation for the discrepancy between experimental work and our corpus results is that, although SSP compliance rates are high in both Russian and English, Russian speakers have more frequent exposure to SSP-violating syllables than English speakers. Even a relatively small difference in exposure to SSP-violating structures may be sufficient to shape speakers' perception of these structures, leading to the behavioral differences observed in experimental studies.needed to determine whether language-wide estimates of SSP prevalence, such as those presented here, fully capture the relevant mechanisms that modulate the role of sonority in speech perception. Further research is needed to determine whether language-wide estimates of SSP prevalence, such as those presented here, fully capture the relevant mechanisms that modulate the role of sonority in speech perception.

Overall, the present study updates prior estimates of cross-linguistic compliance with the SSP, showing 1) a substantially higher presence of SSP-following syllables than SSP-violating syllables across many languages, and that 2) the SSP-following syllables occur more frequently in everyday speech than the SSP-violating ones. It has been proposed that the learning mechanisms underlying language acquisition have shaped the structure of natural languages over time, such that their organization reflects the very abilities humans use to learn them (Chater & Christiansen, 2010; Christiansen & Chater, 2008, 2016; Saffran, 2001, 2002). Our findings are consistent with this view: the syllabic structures most prevalent across language families (SSP-following) are also those that human learners show a preference for from the earliest stages of language development.

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Appendix A

Model structure

Likelihood:

Global prevalence of the SSP:

$$
\begin{aligned}
y_{ij} &\sim \text{Bernoulli } (\frac{\theta_{ij}}{1-\theta_{ij}}) \\
\theta_{ij} &= \alpha + u_j + (\beta + w_j) \cdot \text{Frequency}_{ij}
\end{aligned}
$$

Prior:

$$
\begin{aligned}
\alpha, \beta &\sim \text{Normal } (0, 1) \\
u_j, w_j &\sim \text{Normal } (0, \sigma_{u,w}) \\
\sigma_{u,w} &\sim \text{HalfNormal } (0.5)
\end{aligned}
\tag{1}
$$

Where:

i indexes each observation and j indexes the language it belongs to

y_{ij} indexes whether the syllable follows (1) or violates (0) the SSP

u_j, w_j are language-specific adjustments to α and β

Appendix B

Number of SSP-following and SSP-violating syllables per language

Table B1

Number of available words and syllables available in the dataset, and number and proportion of SSP-following syllables for each language in the dataset.

ISO-639-3	Words	Syllables	SSP+	%	ISO-639-3	Words	Syllables	SSP+	%
cmn	1,076	1,075	1,075	100.0%	san	230	222	216	97.3%
wuu	554	406	406	100.0%	nya	117	135	131	97.0%
kor	4,238	134	134	100.0%	kaz	330	247	239	96.8%
yor	478	304	304	100.0%	epo	1,472	146	141	96.6%
hsn	717	717	717	100.0%	cat	5,446	1,444	1,390	96.3%
mya	460	68	68	100.0%	slv	755	186	179	96.2%
ceb	484	353	353	100.0%	mlt	1,220	860	826	96.0%
kmr	147	139	139	100.0%	fin	9,542	317	304	95.9%
lao	162	142	142	100.0%	hun	4,826	293	280	95.6%
jpn	3,069	69	69	100.0%	lit	1,547	269	255	94.8%
haw	133	67	67	100.0%	afr	344	322	305	94.7%
nan	1,663	856	856	100.0%	eus	2,199	369	349	94.6%
hat	133	73	73	100.0%	swe	670	474	448	94.5%
tlh	316	294	293	99.7%	ell	641	289	271	93.8%
por	14,776	2,642	2,631	99.6%	ita	5,459	1,282	1,197	93.4%
heb	606	459	457	99.6%	mkd	3,036	268	250	93.3%
mar	485	431	429	99.5%	ltz	422	246	228	92.7%
glg	1,112	613	610	99.5%	ces	2,430	231	214	92.6%
hye	2,665	919	913	99.3%	nld	2,430	1,408	1,299	92.3%
kir	113	114	113	99.1%	ukr	2,271	668	615	92.1%
khm	662	397	393	99.0%	dan	902	489	450	92.0%
arz	191	173	171	98.8%	cym	917	276	253	91.7%
spa	15,286	1,598	1,578	98.7%	eng	9,783	3,823	3,489	91.3%
urd	304	306	302	98.7%	deu	2,511	975	888	91.1%
pan	443	302	298	98.7%	fra	5,969	2,100	1,909	90.9%
tur	585	593	585	98.7%	pol	6,047	2,495	2,230	89.4%
hin	3,521	1,647	1,624	98.6%	slk	2,442	187	167	89.3%
tgl	3,899	1,547	1,523	98.4%	bod	223	236	210	89.0%
gle	1,392	477	469	98.3%	rus	19,593	1,702	1,505	88.4%
ben	1,531	243	238	97.9%	bel	623	211	184	87.2%
gla	933	397	388	97.7%	hbs	3,314	312	267	85.6%
ind	3,368	1,041	1,015	97.5%	isl	1,193	292	219	75.0%
ron	1,034	608	592	97.4%	vie	1,913	1,026	767	74.8%
tgk	269	225	219	97.3%	-	-	-	-	-

Appendix C

Results after classifying flat sonority contours as SSP-violating

Figure C1

Observed (crosses) and model-predicted (circles) estimates of SSP prevalence across languages in the dataset. Intervals indicate the 95% and 79% CrI of the posterior distribution. The number of syllables included in the dataset is indicated next to each language.

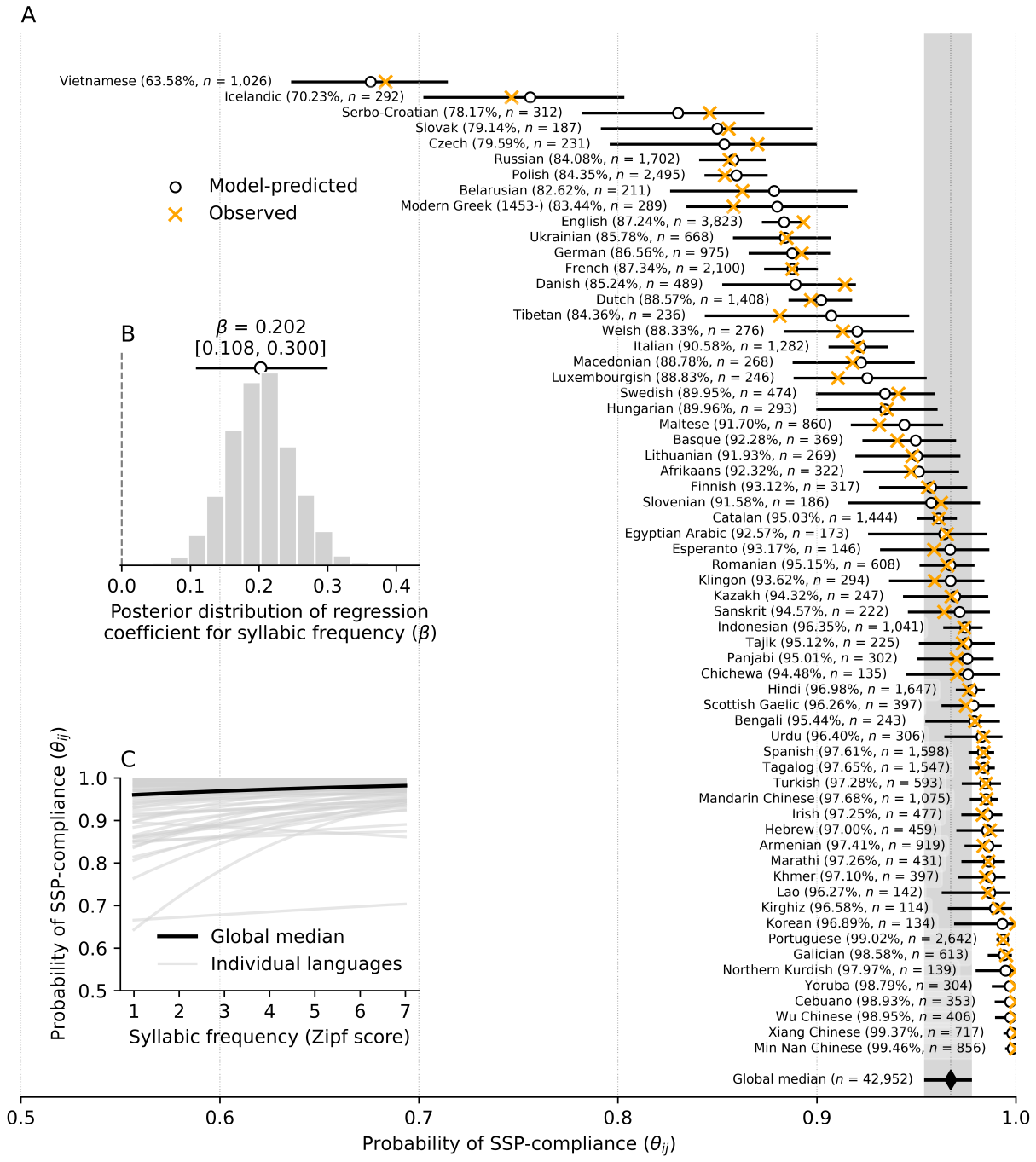


Figure C2

Mean predictions of SSP compliance as a function of syllabic frequency. Predicted cross-language probability of SSP compliance ($\theta_i j$) is shown as a black line. Thin lines indicate language-specific predictions, color-coded by language macro-family.

